

ORM/Entity Framework

Laborator 5

Tehnici avansate de programare

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Code quality metrics

- Software Improvement Group - Maintainability: guidance for producers
- <https://softwareimprovementgroup.com/wp-content/uploads/SIG-TUViT-Evaluation-Criteria-Trusted-Product-Maintainability-Guidance-for-producers.pdf>

Upcoming developer Events

- <https://build.microsoft.com/en-US/home>
 - June 2–3, 2026 / San Francisco and online

Infrastructure layer

- Implement the infrastructure persistence layer with Entity Framework Core

Videos

- <https://learn.microsoft.com/en-us/shows/entity-framework-core-for-beginners/getting-started-with-entity-framework-core/>
- <https://learn.microsoft.com/en-us/shows/entity-framework-core-for-beginners/>
- <https://learn.microsoft.com/en-us/shows/visual-studio-toolbox/entity-framework-core-part-1>
- <https://learn.microsoft.com/en-us/shows/visual-studio-toolbox/entity-framework-core-in-depth-part-1>



Learning sources

- <https://learn.microsoft.com/en-us/azure/architecture/guide/design-principles/scale-out>
- <https://learn.microsoft.com/en-us/azure/well-architected/reliability/principles>
- <https://www.ibm.com/think/topics/data-integrity>
- <https://learn.microsoft.com/en-us/training/paths/get-started-querying-with-transact-sql/>
- <https://www.mongodb.com/resources/basics/databases/acid-transactions>

Exercises

- <https://learn.microsoft.com/en-us/ef/core/get-started/overview/first-app>

Databases

- SqlServer SqlExpress LocalDb
- SqlServer
- Azure SQL Server



Mandatory homework III

- Functional requirement: OnlineShopping application
 - Add/search products
 - Handle shopping cart
 - Checkout order
- Technical requirements for EF Core
 - Repository pattern
 - Model configurations
 - CRUD operations
 - Loading related data
 - Database migrations
 - Database transactions
- Deadline: 30 March 2026 end of the day